

Saul Cooperman

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Summary

Software Engineer focused on building reliable, high-performance distributed systems in C++ and Python. Currently developing latency-sensitive middleware at Bloomberg L.P., supporting thousands of real-time services. Strong background in multithreading, distributed systems, and cross-team engineering.

Experience

- Bloomberg L.P.**, Software Engineer London, UK
 Sept 2024 – present
- Core maintainer of a proprietary distributed messaging platform and gateway infrastructure processing 300B+ messages daily with 99.999% uptime, serving as mission-critical infrastructure across the firm.
 - Lead development of high-performance, multithreaded C++ and Python SDKs with native bindings used by 17,000+ internal microservices across diverse production environments.
 - Increased CI reliability by 30% by building tooling to diagnose flaky failures, leading performance optimization efforts for a high-performance Python library with Cython bindings, and establishing automated benchmarking infrastructure to detect regressions across library releases.
 - Architect and own a schema-driven Python code generation framework adopted by 3,000+ production services (up from ~700), enabling versioned package distribution at enterprise scale.
- Bloomberg L.P.**, Software Engineer Intern London, UK
 June 2023 – Sept 2023
- Built a system to analyse live encoded requests and evaluate backward-incompatible changes to the schema which encoded them.
 - Enabled teams to evolve message schemas comfortably.

Education

- BSc University of Leeds**, Computer Science Leeds, UK
 Sept 2021 – July 2024
- First Class Honours.
 - Final Year Project (First): A neuromechanical model of *C. elegans* steering behaviour integrating sensory and neural mechanisms, optimized using evolutionary algorithms.
- A-Level City of London School**, Mathematics, Further Mathematics, Physics, Chemistry London, UK
 Sept 2019 – July 2021
- Mathematics/Further Mathematics **D1 (A**)**.
- Pre-U** • Awarded “The Worshipful Company of Needle-maker’s Prize for IT and Computing”.

Personal Projects

- PyStack subinterpreter support (open-source Python debugging tool)**
- Contributed subinterpreter support, enabling accurate reconstruction of both Python and native (C) stack frames across multiple subinterpreters within a single process.
- Generic Compilation Database Generator**
- Reimplemented the C++ tool Bear in Rust, generating compilation database for Clang tooling by intercepting Linux build processes via LD_PRELOAD, syscall hooking, and a Unix Domain Socket server with Protobuf for structured data exchange.
- TFL Times - Real-time Transit App**
- Built a React frontend backed by a Python asyncio WebSocket server, including a fully custom typed TTL caching layer with extensive test coverage to minimise external API calls and latency.
- Overengineered Tic-Tac-Toe**
- Implemented a Tic-Tac-Toe engine in C++ with Python bindings via pybind11, using bitboards.
 - Explored ARM NEON SIMD intrinsics for board evaluation and low-level perf optimizations.
- Open source contributions to Pystack, Cython, CPython, BDE**